

University of Texas at Dallas
Department of Electrical Engineering
EEDG/CE 6306 - Application Specific Integrated Circuit Design
Homework #6

Submission:

(a) Written Report (PDF)

- **MSDAP report**
- **Physical Design questions**

(b) Final Verilog source code

(c) Genus scripts and reports

(d) Innovus scripts and reports

Submit onto eLearning a zip file (team): <First name #1>_<Last name #1>_<First name #2>_<Last name #2>_HW06.zip

* If you have any question about the homework, you may contact TA Lucas Wick

Purpose: The goal of this assignment is to complete the initial physical implementation of the MSDAP ASIC in ASAP7 library cells (on No Machine) and memory macros (provided on eLearning). You are given a starting floorplan in eLearning and you will implement the physical design using Cadence Innovus. Evaluate the power, performance and area.

You can continue on working on or make changes to your RTL code and specification documents. Include the Final Verilog source code with your final project, even if it is not changed from the midterm or HW#5. If the RTL is changed, re-run synthesis and include the Genus reports in the final project report. This should be the "Final Data-in" version of the Verilog meaning the Verilog is frozen, but you might be modifying your Genus and Innovus scripts up until the final project. Upload your Genus and Innovus scripts and reports.

If you had difficulty completing the top-level design of MSDAP in HW#5, you can use the solution RTL in eLearning. This RTL does not have the memories integrated, so you will need to do so.

Tutorials of Genus and Innovus, and related files are posted on eLearning.

(0) Top-level gate simulation

(0.1) Run top-level gate-level netlist simulations of your synthesis netlist, compare to the RTL simulations and explain why your simulations prove your design is correct

(1) Physical Design of MSDAP using ASAP7

(1.1) Correctly generate the initial layout by Innovus and report your timing violations, DRC and LVS errors. You do not need to be Timing or PV clean for HW#5, but you do need to be Timing and PV clean for the final project (excluding known DRC errors). Include the initial Innovus report in your submission.

(1.2) In the project report, analyze the initial performance of your MSDAP with following aspects covered:

(1.2.1) Performance: Frequency of Sclk

(1.2.2) Power consumption: Switching and Leakage

(1.2.3) Area: Report of Number of Cells

(1.2.3) Chip Area: in mm²

(1.3) Include a picture of your floorplan in the written report

(1.4) Provide a discussion about how will you achieve Timing clean and PV clean for the final report

(1.5) If you used the solution RTL for the MSDAP, compare and contrast your design/microarchitecture with the given solution to get the full credit.